D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Recent advances in supportive care in cancer chemotherapy.

February 2022 | DM 0222 | Q2 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction — supportive care as anticancer therapy

Supportive care prevents and treats complications of cancer and its treatment — enabling delivery of full-dose intensive regimens, preserving organ function, and improving quality of life and sometimes survival. 'Recent advances' for a 2022-onward Paper IV answer span antiemetics (olanzapine), growth factors, bone-modifying agents, cardio-oncology, immuno-toxicity algorithms, nutrition, VTE prevention, and digital PRO monitoring.

Structure: CINV → FN/G-CSF → anemia/ESA caution → bone health → cardio-oncology → irAEs → mucositis/nutrition → VTE → CIPN/fatigue/fertility → NCCN 2026 supportive bullets → Indian practice.

- Supportive care is not optional add-on — it is part of the treatment protocol.
- Cite MASCC/ESMO/ASCO antiemetic guidelines and NCCN Supportive Care libraries.

EXAM PEARLS / HALLMARKS

- Start essay with: 'Cure rates rise when toxicity is anticipated and protocolized.'

2. CINV — modern risk-adapted antiemesis

Chemotherapy-induced nausea/vomiting is classified as acute (≤24 h), delayed (>24 h), anticipatory, breakthrough, and refractory. Emetogenicity tiers (high/moderate/low/minimal) drive prophylaxis.

Cornerstone classes: 5-HT3 antagonists (ondansetron, granisetron, palonosetron), NK1 receptor antagonists (aprepitant/fosaprepitant/netupitant), dexamethasone, and **olanzapine** — now a practice-changing addition for HEC and breakthrough.

- HEC (e.g., cisplatin, AC): 4-drug prophylaxis often used — 5-HT3 + NK1 + dexamethasone + olanzapine.
- Palonosetron: longer half-life; useful for delayed CINV.
- Non-pharmacologic: hydration, ginger evidence modest, behavioral therapy for anticipatory CINV.

Peripheral 5-HT3	Acute CINV — enterochromaffin → vagus
Central NK1 / substance P	Delayed CINV
Dopamine / multiple receptors	Olanzapine broad antiemetic cover
Cortex / memory	Anticipatory CINV

Figure: mechanism / pathway map

3. Olanzapine evidence & practical dosing

Navari et al. and subsequent trials showed olanzapine improves nausea control — the endpoint patients care about most — when added to triple therapy for HEC. Typical investigational/clinical dose: **olanzapine 5–10 mg PO nightly** on days 1–4 (5 mg often preferred in elderly to reduce sedation).

Breakthrough: olanzapine is a preferred agent in many guidelines when not already used for prophylaxis.

Trial	Key result / exam takeaway
Navari NEJM / JCO olanzapine trials	Olanzapine ↑ no-nausea rates with HEC triple therapy
MASCC/ESMO & ASCO antiemetic updates	Olanzapine incorporated into HEC pathways

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Olanzapine (HEC prophylaxis)	5–10 mg PO days 1–4 (prefer 5 mg if sedation risk)
Aprepitant	125 mg d1 → 80 mg d2–3 (or fosaprepitant 150 mg IV d1)
Ondansetron	8 mg IV/PO schedule per regimen; watch QT
Dexamethasone (with NK1)	12 mg d1 then tapered per guideline (NK1 interaction)
Palonosetron	0.25 mg IV day 1

Table: doses and regimens

EXAM PEARLS / HALLMARKS

- Warn about sedation and metabolic effects; avoid co-administration issues with CNS depressants.

4. Febrile neutropenia — prevention & treatment advances

FN remains a medical emergency: temperature ≥38.3°C (or ≥38.0°C sustained) + ANC <500/µL (or <1000 with predicted decline). Risk tools: ASCO/NCCN regimen risk (≥20% FN → primary G-CSF), MASCC/CISNE scores for outpatient eligibility after careful selection.

Advances: biosimilar filgrastim/pegfilgrastim improving access; same-day vs next-day pegfilgrastim debates; antibiotic stewardship with early de-escalation when cultures negative and recovering.

- High-risk FN: hypotension, pneumonia, uncontrolled cancer, ANC <100, duration >7 d expected.
- Never delay first antibiotic dose for imaging.

FN prevention and acute management

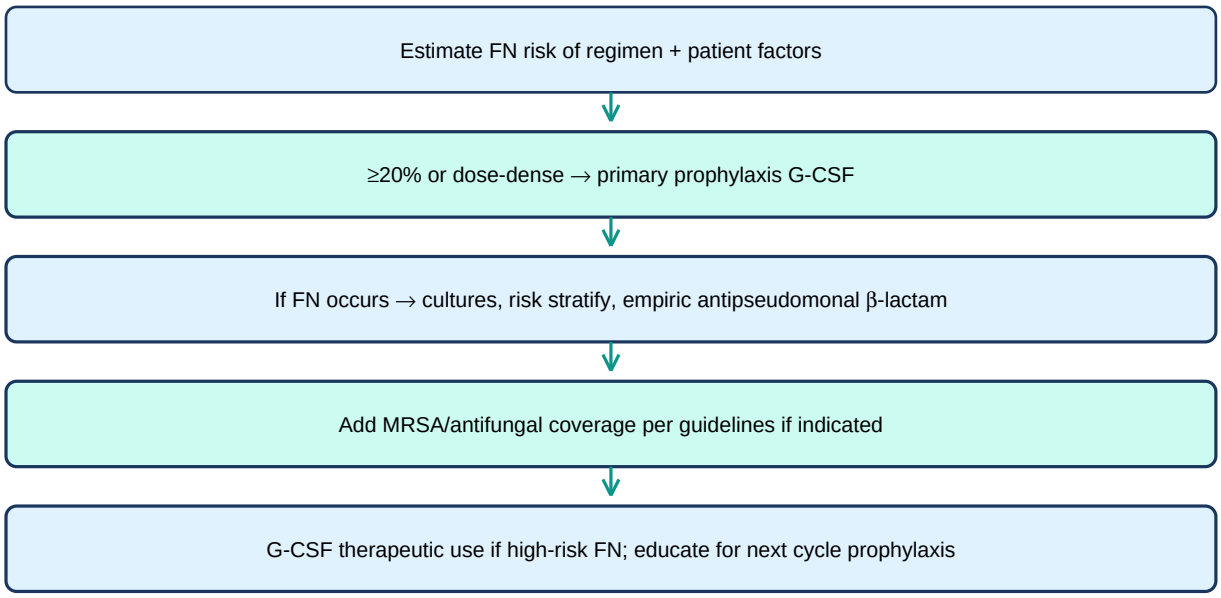


Figure: stepwise exam algorithm

5. G-CSF / GM-CSF — doses & pearls

Filgrastim 5 µg/kg SC daily until ANC recovery (start 24–72 h post chemo). Pegfilgrastim 6 mg SC once per cycle (≥24 h after chemo; avoid within 14 days before next chemo). Self-administration and on-body injectors are recent practical advances. Biosimilars are preferred cost-saving options in Indian practice.

- Bone pain: manage with NSAIDs/loratadine as used in practice pathways.
- Do not use G-CSF concurrently with chemo/RT in most protocols (except specific regimens).

Drug / regimen	Dose / schedule (exam pearls)
Filgrastim	5 µg/kg SC daily; start ≥24 h after myelosuppressive chemo
Pegfilgrastim	6 mg SC once per cycle
Trilaciclib (selected ES-SCLC)	Myelopreservation before chemo — niche advance

Table: doses and regimens

6. Anemia, ESAs & transfusion — caution first

Cancer- and chemo-related anemia: correct iron deficiency (IV iron when appropriate), B12/folate, bleed, hemolysis. Transfuse based on symptoms and comorbidities, not a fixed Hb alone.

ESA caution: erythropoiesis-stimulating agents may shorten survival / promote tumor progression in some settings and ↑ VTE. Restrict to selected chemo-induced anemia with non-curative intent under label/REMS-like caution; avoid in curative pathways when possible.

- Target lowest Hb to avoid transfusion — do not normalize Hb with ESA.
- IV iron formulations (ferric carboxymaltose etc.) useful if absolute/functional iron deficiency.
- Exam phrase: ‘ESA — symptom control in palliative chemo anemia only, after risk discussion.’

EXAM PEARLS / HALLMARKS

- FDA boxed warnings on ESAs are classic viva fodder.

7. Bone-modifying agents — SRE prevention

Zoledronic acid and denosumab reduce skeletal-related events in bone metastases (breast, prostate, others) and treat hypercalcemia of malignancy. Denosumab (RANKL mAb) delayed SREs vs zoledronic acid in some solid-tumor trials; choose by renal function, cost, and ONJ risk counseling.

Adjuvant bisphosphonates in postmenopausal early breast cancer reduce bone recurrence and improve OS in meta-analyses — mention as supportive/adjuvant overlap.

- ONJ risk: dental review, avoid invasive dental procedures on therapy when possible.
- Hypocalcemia: especially denosumab — monitor Ca/Mg.
- Renal dosing: adjust/hold zoledronic acid for CrCl limits; denosumab not renally cleared the same way but watch Ca.

Drug / regimen	Dose / schedule (exam pearls)
Zoledronic acid (bone mets)	4 mg IV q4–12 weeks (extend interval if stable)
Denosumab (bone mets)	120 mg SC q4 weeks
Zoledronic acid (hypercalcemia)	4 mg IV; dental check when feasible
Calcium + vitamin D	Supplement with denosumab/bisphosphonate unless hypercalcemic

Table: doses and regimens

8. Cardio-oncology advances

Anthracyclines (cumulative dose, liposomal formulations), HER2 agents, VEGF TKIs/bevacizumab (hypertension, ischemia), ibrutinib (AF), and checkpoint myocarditis define modern cardio-oncology. Advances: baseline risk stratification, echo/GLS, troponin surveillance in high-risk protocols, aggressive BP control on VEGF inhibitors, and rapid steroid pathways for IO myocarditis.

- Hold HER2 therapy for significant LVEF drops per institutional/ASCO rules; cardio referral.
- IO myocarditis: rare but high mortality — early ECG/troponin, pulse steroids, discontinue CPI.
- Dexrazoxane in selected high cumulative anthracycline settings (guidelines vary).

Cardio-oncology loop

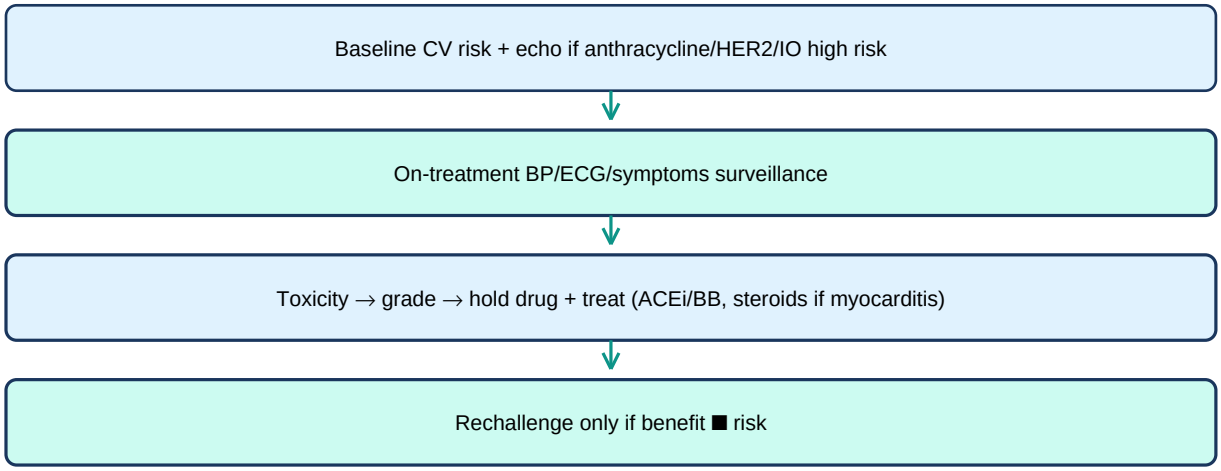


Figure: stepwise exam algorithm

9. Immune-related adverse events (irAEs) — IRIS-adjacent care

Checkpoint inhibitor expansion made irAE management a core supportive-care competency. CTCAE grading drives hold/steroid decisions. Common: dermatitis, colitis, hepatitis, endocrinopathies (thyroid, hypophysitis, adrenal), pneumonitis; rare: myocarditis, myasthenia, severe neurologic disease.

'IRIS' in exam answers may be framed as immune reconstitution-type flares in HIV or as immune-related inflammatory syndromes after CPI — clarify terminology and discuss steroid-responsive inflammation with multidisciplinary care.

- Grade 2: usually hold CPI + start corticosteroids (e.g., prednisone 0.5–1 mg/kg).
- Grade 3–4: high-dose steroids; infliximab for steroid-refractory colitis; MMF for hepatitis; permanent discontinuation for many grade 4.
- Endocrinopathies: hormone replacement; CPI often continued once stable on replacement.

Drug / regimen	Dose / schedule (exam pearls)
Prednisone (gr 2 irAE)	~0.5–1 mg/kg/day with taper
Methylprednisolone (severe)	1–2 mg/kg/day IV
Infliximab (refractory colitis)	5 mg/kg IV — after excluding infection

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* NCCN Management of Immunotherapy-Related Toxicities (current 2026 family); Kidney Cancer v2.2027 (IO combinations context)
- Follow NCCN immunotherapy toxicity algorithms for hold/discontinue/steroid/infliximab pathways.
- Educate patients with wallet cards — early reporting reduces fatalities from myocarditis/pneumonitis.

10. Mucositis, nutrition & cachexia

Oral mucositis: oral cryotherapy for bolus 5-FU, dental prep, magic mouthwash variants, palifermin in selected transplant TBI regimens, photobiomodulation in some centers. Nutrition: early dietitian referral; treat reversible causes; consider enteral feeding before parenteral. Cachexia: multimodal — exercise, nutrition, treat inflammation; ghrelin mimetics/other agents evolving.

- NCCN Distress Thermometer + nutrition screening at key visits.
- Avoid alcohol-based mouth rinses that worsen mucositis.
- Tube feeding preferred over TPN if gut works.

11. VTE prevention & treatment in cancer

Cancer VTE risk: tumor type (pancreas, stomach, brain), stage, chemo (platinum, IMiDs + steroids), central lines. Khorana score guides primary prophylaxis in ambulatory high-risk patients.

Advances: DOACs (apixaban, rivaroxaban, edoxaban) validated vs LMWH in selected cancer VTE (select GI/GU mucosal bleed risk caution). LMWH remains preferred in many GI luminal cancers or very high bleed risk.

- Primary prophylaxis: consider in Khorana high-risk ambulatory outpatients after bleed assessment.
- IMiDs in myeloma: aspirin/LMWH/DOAC thromboprophylaxis per risk.

Drug / regimen	Dose / schedule (exam pearls)
Enoxaparin (treatment)	1 mg/kg SC BID or 1.5 mg/kg daily (adjust CrCl)
Apixaban (selected cancer VTE)	10 mg BID ×7 d → 5 mg BID (trial protocols)
Rivaroxaban (selected)	15 mg BID ×21 d → 20 mg daily with food

Table: doses and regimens

12. CIPN, fatigue, fertility, survivorship

Chemotherapy-induced peripheral neuropathy (taxanes, platinum, vinca, bortezomib): dose holds/reductions; duloxetine has best evidence for painful CIPN. Cryotherapy/compression for taxanes — supportive data evolving.

Fatigue: treat contributors (anemia, thyroid, depression, sleep); structured exercise is an evidence-based intervention — now even embedded in some NCCN disease guidelines (e.g., post-adjuvant CRC exercise).

Fertility: sperm cryopreservation; oocyte/embryo freezing; GnRH analogs during chemo in selected breast cancer — discuss before gonadotoxic therapy.

- Vaccinations: inactivated vaccines timing; COVID/influenza; avoid live vaccines while immunosuppressed.
- Psychosocial oncology and palliative integration early in metastatic disease.

13. Growth factors for platelets & special agents

Romiplostim/eltrombopag: used in selected chemo-induced thrombocytopenia pathways and bone marrow failure — specialist use. TPO receptor agonists appear in modern supportive algorithms when dose intensity is critical and plateletdelay is the bottleneck.

- Not a substitute for treating underlying marrow disease.
- Watch thrombosis risk with TPO-RAs.

14. NCCN 2026 supportive-care relevant updates

Even when answering a classic ‘supportive care in chemotherapy’ essay, inject **NCCN 2026** currency: exercise prescription, CVD risk pages, and digital navigation tools show examiners you are up to date.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* NCCN Supportive Care guidelines family (antiemesis, myeloid growth factors, anemia, survivorship, immunotherapy toxicity); Colon Cancer v3.2026 (exercise); Prostate Cancer v1.2027 (CVD risk assessment page)
- Structured exercise ≥150 min moderate–vigorous activity/week after adjuvant CRC therapy — formal NCCN recommendation.
- Prostate NCCN: CVD risk-assessment page added — model for cardio-oncology documentation.
- NCCN Guidelines Navigator digital tool expanding — use for rapid antiemesis/growth-factor look-ups.
- Immunotherapy toxicity NCCN pathways are mandatory knowledge as chemo-IO combinations proliferate.

15. Indian / NCG practical supportive care

Biosimilar G-CSF and antiemetics improved access. Challenges: dental ONJ prevention logistics, echo wait times, DOAC cost vs LMWH, and uneven dietitian/palliative staffing. NCG centers emphasize protocolized FN pathways and antimicrobial stewardship.

- Teach patients FN precautions in local language; provide 24h contact number.
- Olanzapine is inexpensive and high-value — highlight for Indian exams.
- Avoid unnecessary ESA use in curative breast/lymphoma pathways.

16. Summary flowchart & pearls

Recent advances make supportive care proactive and evidence-based: olanzapine-containing antiemesis, risk-adapted G-CSF, cautious ESA use, optimized bone agents, cardio-oncology surveillance, irAE algorithms, DOAC options for VTE, and lifestyle prescriptions.

Supportive care across the treatment continuum

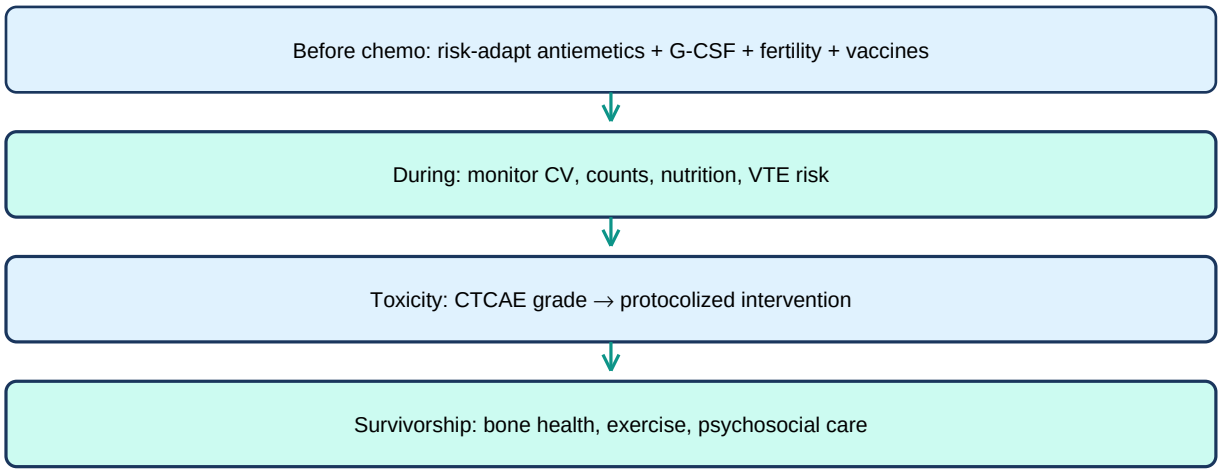


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- HEC: remember olanzapine 5–10 mg d1–4.
- FN ≥20% risk → primary pegfilgrastim/filgrastim.
- ESA: non-curative + informed consent only.
- Denosumab 120 mg SC q4w vs zoledronic acid 4 mg IV — ONJ counseling both.
- irAE grade 3 colitis: steroids → infliximab if refractory.
- Exercise is now a guideline-backed anticancer adjunct.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

Work-up algorithm (draw this in the answer booklet)

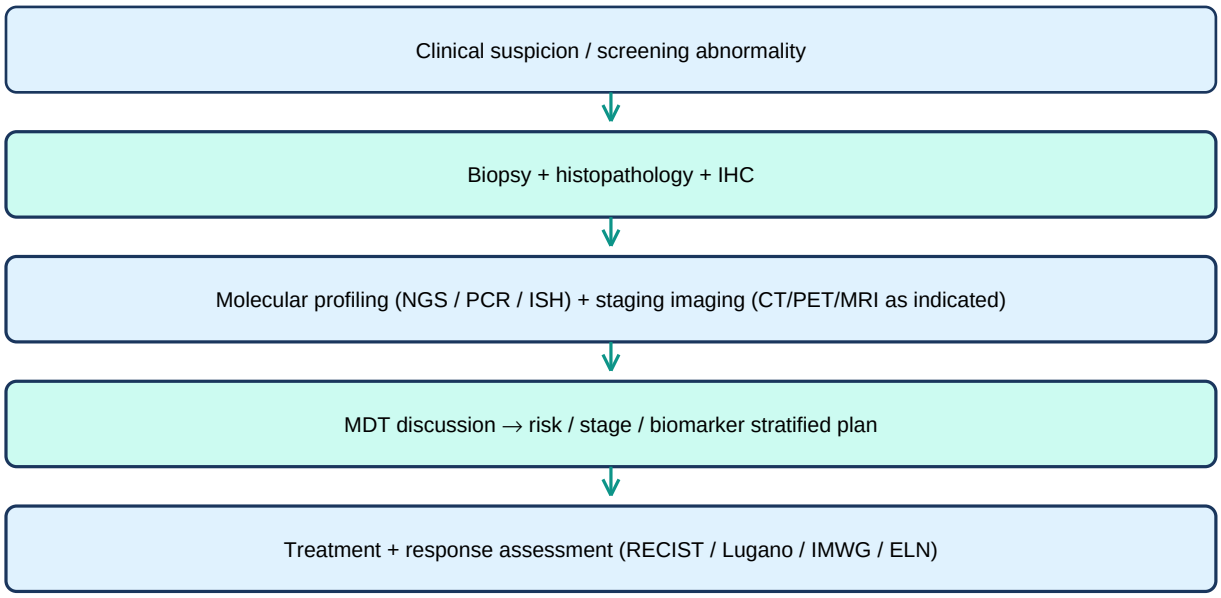


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

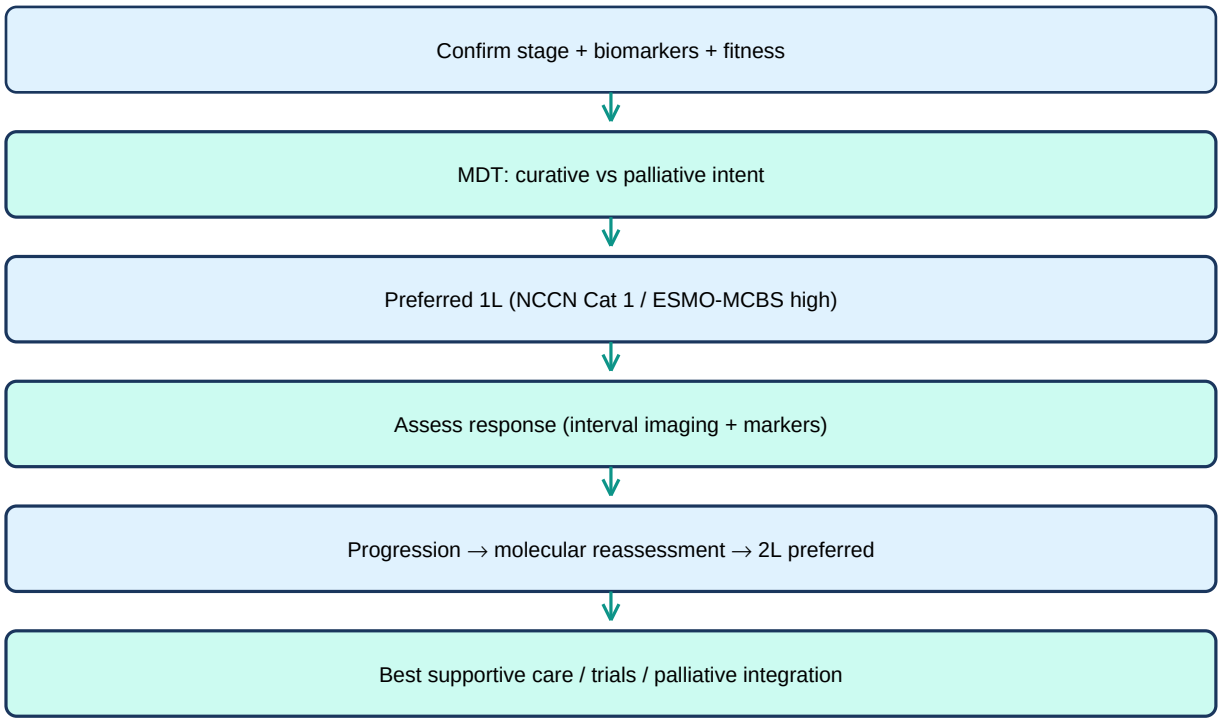


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

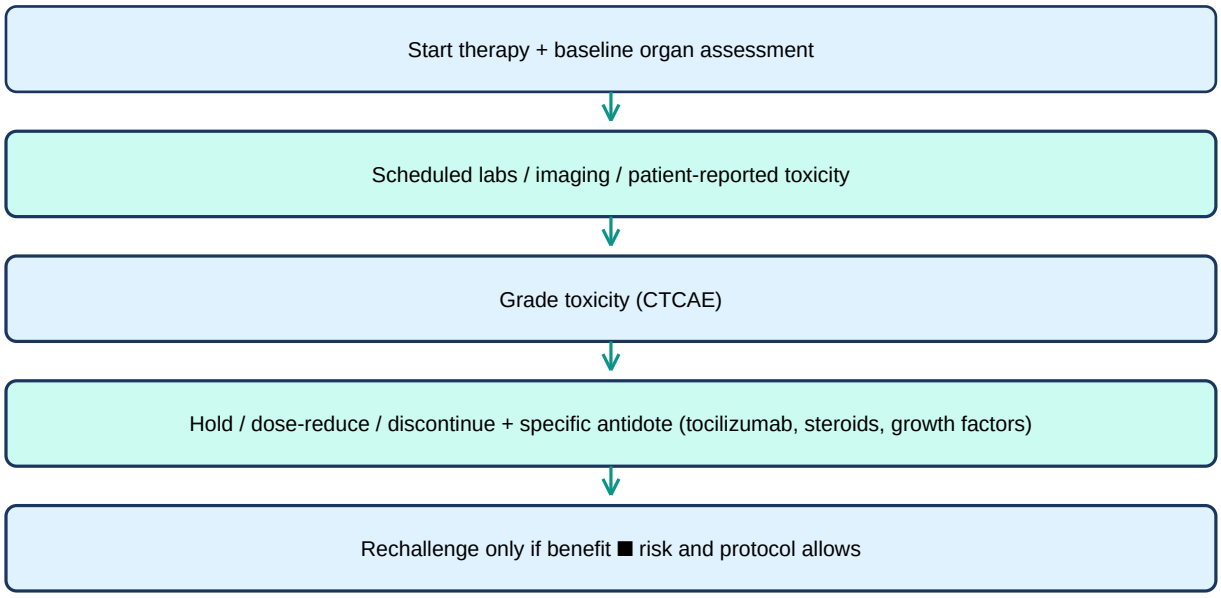


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* NCCN Guidelines® (latest 2026/2027 disease-specific version)
- Cite the latest disease-specific NCCN Guidelines® (2026/2027 versions) and NCCN Drugs & Biologics Compendium for category ratings.
- Mention Category 1 vs 2A/2B and preferred vs other recommended vs useful in certain circumstances.
- Cross-check FDA label, institutional availability (India), and National Cancer Grid pathways where relevant.

EXAM PEARLS / HALLMARKS

- NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).
- Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

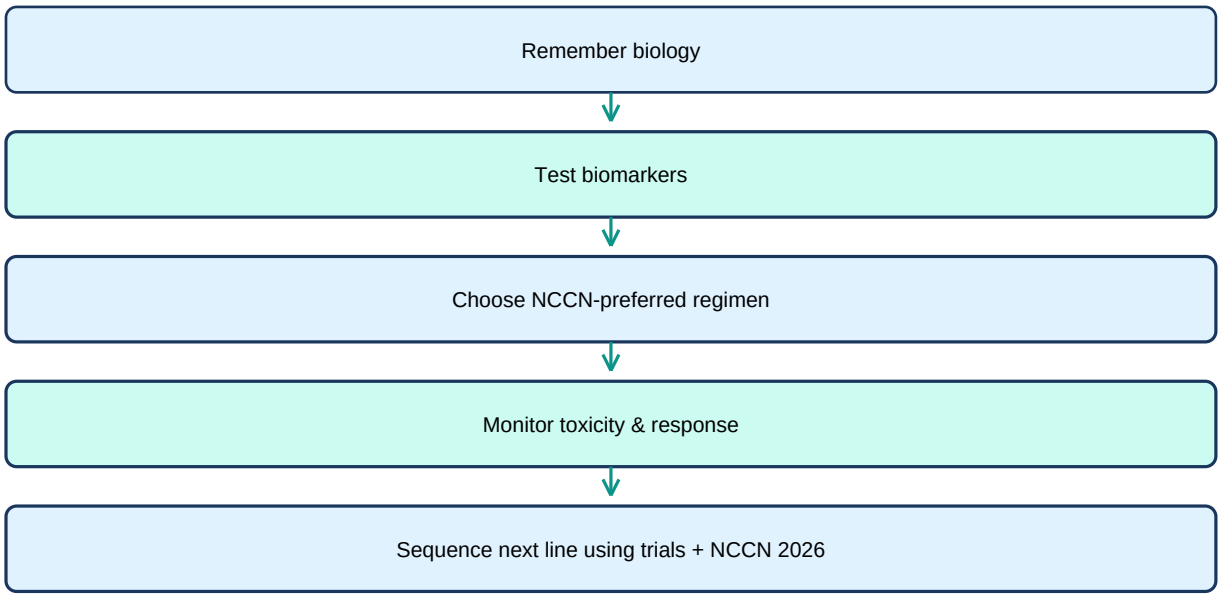


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

• ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

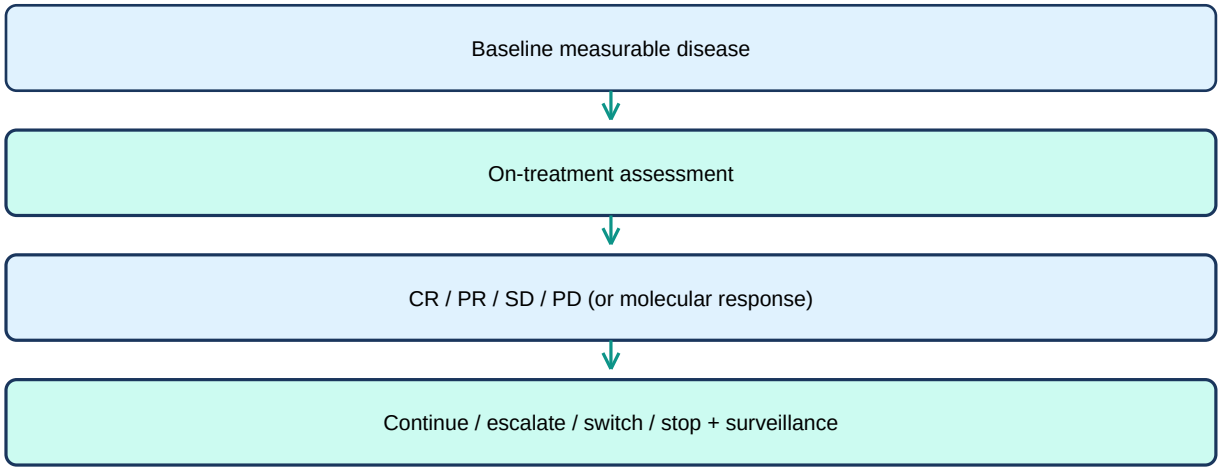


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

• Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.

• Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

• Document MDT decisions; clinical trial referral when standard options exhausted.

• Counsel financial toxicity transparently; prioritize highest-value Category 1 options.

• Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

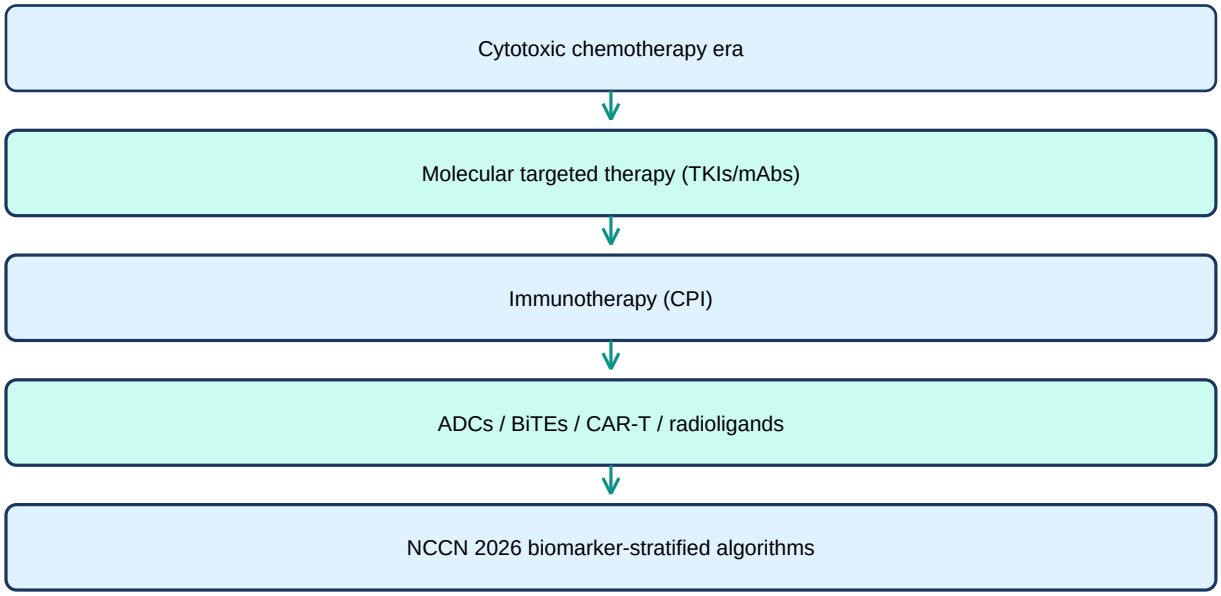


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

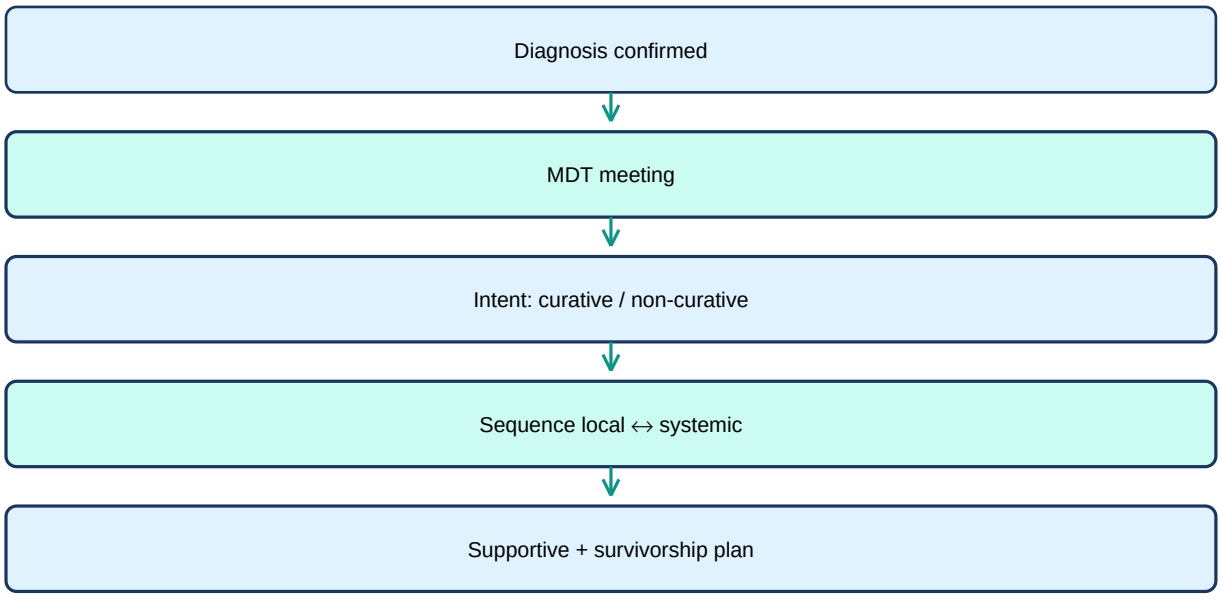


Figure: stepwise exam algorithm

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome

in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

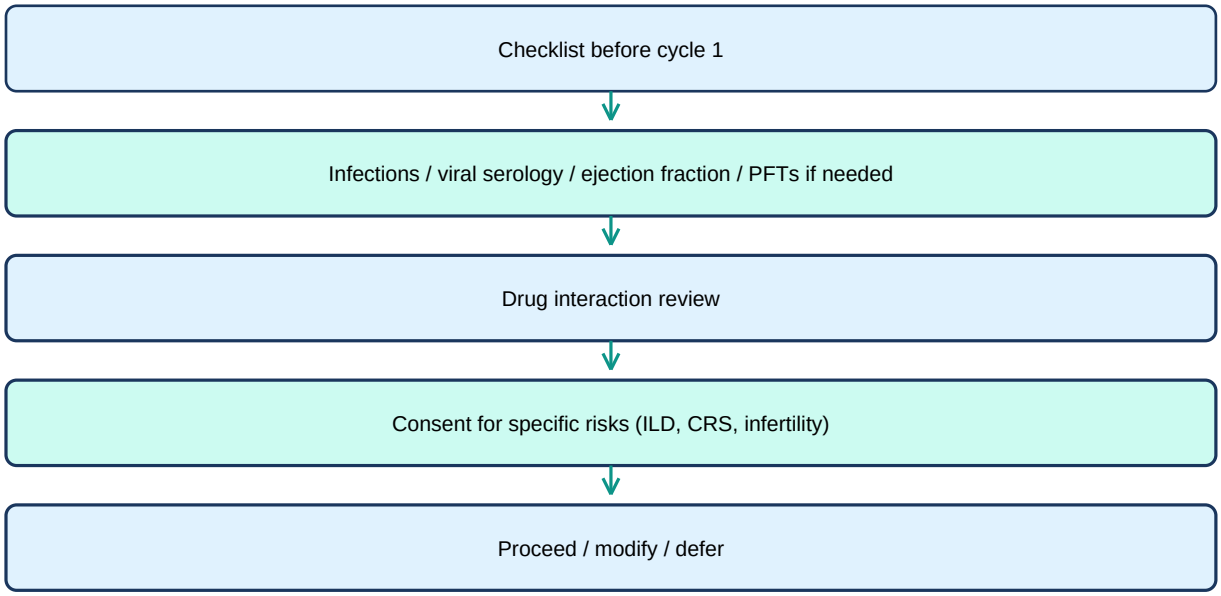


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write 'NCCN 2026' in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
HERA / BCIRG-006 / NSABP B-31 / N9831	Adjuvant trastuzumab OS/DFS benefit
CLEOPATRA	Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MOSAIC / C-07 / IDEA	Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX
ATOMIC	FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred
ZUMA-1 / JULIET / TRANSCEND	Axi-cel; tisa-cel; liso-cel LBCL
ZUMA-7 / TRANSFORM / BELINDA	2L CAR-T vs SOC

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Nivolumab	240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen
Olaparib	300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year
Trastuzumab	8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose)

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* NCCN Guidelines® (latest 2026/2027 disease-specific version)

• Cite the latest disease-specific NCCN Guidelines® (2026/2027 versions) and NCCN Drugs & Biologics Compendium for category ratings.

• Mention Category 1 vs 2A/2B and preferred vs other recommended vs useful in certain circumstances.

• Cross-check FDA label, institutional availability (India), and National Cancer Grid pathways where relevant.

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

• If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.

• Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

NCCN 2026 / 2027 UPDATES (include in answer)

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.